

Elimination matrices

I believe we now possess the skills to think of row reduction on matrices in a different way.

suppose we have

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$$

and we want to find A^{-1}

The way we think of this is to set up the augmented matrix and perform row reduction

$$[A | I] \sim [I | A^{-1}]$$

I would like to introduce a new idea, the idea of elimination matrices.

say we set up the matrix A :

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$$

Our first step of elimination would be

$$R_2 \leftarrow R_2 - 2R_1$$

The elimination matrix E_1 to perform this operation would be

$$E_1 = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} \begin{array}{l} \text{take } 1R_1 \text{ and } 0R_2 \\ \text{take } -2R_1 \text{ and } 1R_2 \end{array} = \begin{bmatrix} 1 & 3 & 3 \\ -2+2 & -6+7 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$$

The next step would be

$$R_1 \leftarrow R_1 - 3R_2$$

$$E_2 = \begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1-3\cdot 0 & 3-3 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Multiplying E_1 and E_2 in the right order will give us A^{-1} .

We can use the product matrix to do the reduction in one go.

$$E_2 E_1 = \begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} 7 & -3 \\ -2 & 1 \end{bmatrix} = A^{-1}$$